Hardware-Constrained Architecture Search for Lane-Change Intention Prediction from Vehicle Time Series

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***Abstract*—Predicting a driver's lane-change intention early enough to act on it is a time-series classification problem with hard deployment limits: the model has to run on automotive-grade microcontrollers within tight memory and latency budgets. We apply constrained neural architecture search to the Lane Change Intention Recognition driving dataset [3] (windows of 50 samples over 31 signals) and obtain compact one-dimen- sional convolutional networks for three tasks: three-class intention classification and time-to-lane-change regression for left and right maneuvers. [TODO: results: accuracy, RMSE, model size, latency on STM32] The searched models improve on the published base-line (Transformer, 0.51 s time-to-lane-change RMSE, deployed in FP32 [1]) while using [TODO: X]× less memory, and we report measured, not estimated, footprint and latency on the same STM32 targets after int8 quantization.**

***Index Terms*—Neural architecture search, lane-change intention, time-series classification, TinyML, embedded inference, STM32.**

I. Introduction

[TODO: Motivation: intention prediction as advance warning for ADAS; why on-device inference (latency, privacy, cost, no connectivity assumptions); the gap: published DMIR models target accuracy, not deployment budgets.]

Contributions:

1) a constrained search space and objective tuned to one-dimensional driving signals under microcontroller budgets;
2) models that beat the published DMIR baseline on all three tasks at a fraction of its footprint [TODO: verify once results exist];
3) measured deployment numbers (flash, RAM, latency) on [TODO: STM32 part], with the full pipeline released as open source.

II. Related Work

A. Lane-change intention prediction

[TODO: The LC dataset and framework baseline [1] (TTLC regression, Transformer RMSE 0.51 s, FP32 deployment on STM32H7B3/F401); earlier HighD-based TTLC work (RMSE 0.63–0.7); MicroNAS for time series and TinyTNAS as closest NAS relatives.]

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B. Constrained architecture search for microcontrollers

[TODO: muNAS [2], MCUNet/TinyNAS, MicroNets; what transfers to 1D time series and what does not.]

III. Data

We use the Lane Change Intention Recognition dataset [3] (50 human drivers in the CARLA simulator, 10 Hz, over 3,400 annotated lane changes; MIT license), prepared as fixed windows of T=50 samples (5 s) over F=31 channels [TODO: confirm channel order and whether channel 31 is fileTime — drop it if so; official feature count is 30]. Classification uses balanced splits of 94,620 / 16,332 / 19,722 windows (train/val/test) over three classes: no intention, right lane change, left lane change. The regression tasks map a window to the time until the lane change, $t \in [0, 4]$ s discretized at 0.1 s, with 39,313 / 7,742 / 8,097 (LCL) and 33,201 / 5,828 / 7,073 (LCR) windows.

The test splits contain isolated spikes (up to 5×10^6) on two channel pairs that never appear in training; we clip validation and test features to the per-channel training range and leave training data untouched. [TODO: confirm the spike origin with the data provider; state how the baseline handled them.]

IV. Method

A. Search space

[TODO: 1D cells: depthwise-separable and standard convolutions, kernel sizes, widths, depth, pooling; operator set restricted to what the ST Edge AI compiler maps efficiently.]

B. Constraints and search algorithm

[TODO: peak RAM, flash after int8 quantization, MACs; aging evolution with constraint handling; search cost budget.]

V. Experiments

A. Setup

[TODO: training schedule, hardware, seeds/repeats, hyperparameters; all runs logged to the released experiments.jsonl.]

B. Results

C. Deployment on STM32

Target platform: STM32H7B3I-DK discovery kit (Cortex-M7 at 280 MHz, 1.4 MB SRAM, 2 MB flash) — the same MCU family used by the baseline framework [1], which deployed FP32 models only. [TODO: toolchain version, int8 quantization accuracy drop, flash / RAM / latency measured on device, comparison with the baseline's Table III.]

TABLE I

TEST-SET COMPARISON. THE PUBLISHED BASELINE [1] REPORTS A SINGLE TIME-TO-LANE-CHANGE REGRESSION (RMSE 0.51 S, TRANSFORMER); THE THREE-TASK FORMULATION NUMBERS ARE INTERNAL REFERENCE RESULTS ON THE SAME PREPARED DATA. [TODO: FILL OUR ROWS FROM LOGS/EXPERIMENTS.JSONL; ADD PARAMS, FLASH, RAM, MACS COLUMNS; CONFIRM INTERNAL-BASELINE PROVENANCE WITH THE TEAM.]

Model	Acc. (%)	RMSE LCR (s)	RMSE LCL (s)
Internal reference (unpublished)	92.0	0.42	0.44
Published Transformer [1]	—	0.51 (single TTLC)	
Ours (searched)	[TODO:]	[TODO:]	[TODO:]

VI. CONCLUSION

[TODO: two or three sentences; no new claims.]

REFERENCES

[1] L. Forneris *et al.*, "A deployment-oriented simulation framework for deep learning-based lane change prediction," *IEEE Signal Process. Lett.*, vol. 33, pp. 136–140, 2026, doi: 10.1109/LSP.2025.3638676.

[2] E. Liberis, Ł. Dudziak, and N. D. Lane, "μNAS: Constrained neural architecture search for microcontrollers," in *Proc. 1st Workshop Mach. Learn. Syst. (EuroMLSys)*, 2021, pp. 70–79.

[3] L. Forneris *et al.*, "Lane change intention recognition dataset," Zenodo, 2025, doi: 10.5281/zenodo.16686054.